

PARAS RANA

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EDUCATION

Indian Institute of Information Technology, Nagpur

Aug 2023 – May 2027

B.Tech, Computer Science and Engineering | CGPA: 7.4/10

Relevant Coursework: Data Structures & Algorithms, Full-Stack Web Development, Object-Oriented Programming, Database Management Systems, RESTful API Design, Operating Systems, Computer Networks.

PROJECTS

Microkernel-Style User-Level Threading Runtime – Systems Programming Project

[GitHub](#)

Tech: C, Linux, POSIX APIs, ucontext, Synchronization Primitives

- Built a cooperative and preemptive **user-level threading library** supporting thread creation, scheduling, context switching, and lifecycle management.
- Implemented a **round-robin scheduler** with ready/blocked queues and efficient context switching without kernel thread overhead.
- Developed **synchronization primitives** including mutexes, semaphores, condition variables, and thread joining mechanisms.
- Designed a modular **kernel-style architecture** separating scheduler, thread control blocks, context management, and synchronization layers.

Vector Database from Scratch – High-Performance Systems Project

[GitHub](#)

Tech: Modern C++, AVX2 SIMD, HNSW, IVF, Product Quantization, KD-Tree

- Built a vector database from scratch in modern C++, implementing **Linear Scan, KD-Tree, IVF, Product Quantization, and HNSW** for exact and approximate nearest-neighbor search.
- Optimized vector search using **AVX2 SIMD** and cache-efficient memory layouts, achieving **3.3× faster search** and **42.7 QPS** on 100K vectors of 1024 dimensions.
- Benchmarked ANN algorithms across dimensions (4–1024), demonstrating KD-Tree's curse of dimensionality and implementing pruning, quantization, and graph-based retrieval strategies.

Collaborative Music Platform – Full-Stack Web Application

[Live](#) | [Demo](#) | [GitHub](#)

Tech: React.js, Node.js, Express.js, WebSockets, PostgreSQL, Redis, TailwindCSS, RESTful APIs, JWT Authentication, Vercel, Render

- Engineered a real-time synchronized listening platform using **WebSockets** supporting **10–15 concurrent users** with a shared song queue, live group chat, and collaborative playlist recommendations, eliminating all manual coordination overhead.
- Integrated **Redis** for session caching and pub/sub messaging to reduce database load and improve real-time event delivery latency; deployed the frontend on **Vercel** and the backend on **Render** for scalable and consistent production environments.
- Designed a **PostgreSQL** relational schema and **RESTful API** layer with **JWT-based authentication** to power profile-based social discovery, enabling users to browse friends' saved songs and playlists, measurably increasing average session duration and content interaction rates.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL, HTML5, CSS3

Backend Frameworks: FastAPI, Node.js, Express.js, LangChain

Frontend & Web: React.js, Next.js, WebSockets, gRPC, Protocol Buffers

Databases: PostgreSQL, MongoDB, Redis, MySQL, Vector Database

DevOps & Infrastructure: Docker, Linux, CI/CD

ACHIEVEMENTS

- Ranked **125th globally** out of **41,000+ participants** in CodeChef Starters 227; attained global rank **283** and All India rank 67 among **35,000+ participants** in LeetCode Biweekly Contest 180.
- Reached Guardian tier (**2100+ rating**) on LeetCode and **4-Star** (1800+ rating) on CodeChef; earned 3-Star C++ and 4-Star DSA badges on HackerRank; solved **1200+ problems** across all platforms.
- Selected as **National Hackathon Finalist** at the competition hosted by IIT Guwahati, competing against teams from top engineering institutions across India.
- Selected as **Finalist** at the hackathon organized by **IIIT Gwalior**, competing against student teams from premier technical institutes across the country.